Soumendu Majee

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Research Interests

Computational Imaging · Efficient Deep Learning · Camera ISP · X-ray Computed Tomography

Education

Ph.D., Electrical and Computer Engineering *June 2021*
Purdue University, IN, USA
Advisors: Prof. Charles A. Bouman and Prof. Gregery T. Buzzard
Thesis: High Speed Imaging via Advanced Modeling
GPA: 4.0/4.0

B.Tech., Electronics and Electrical Communication Engineering *May 2014*
Indian Institute of Technology, Kharagpur, India
GPA: 9.1/10

Skills

Languages: Python, C++, C, C#, MATLAB, Cython, Bash
ML Frameworks: PyTorch, TensorFlow, Keras, JAX, SNPE
Tools: CUDA, OpenCV, LaTeX

Experience

Staff Researcher II *September 2025 – Present*
Staff Research Engineer *March 2024 – August 2025*
Senior Engineer, Research *April 2022 – February 2024*
MPI Lab, Samsung Research America, TX, USA

- Led the development and commercialization of Unified-Tone-Map across multiple Galaxy camera pipelines for Samsung Galaxy S25 and S26
- Led the development and commercialization of Under Display Camera pipeline for Samsung Galaxy Z Fold5 and Fold6
- Contributed to the development and commercialization of Expert-Raw for Samsung Galaxy S23 and S26
- Contributed to the development and commercialization of Tetra AI Zoom for Samsung Galaxy S24
- Contributed to the development and commercialization of Tetra High-resolution Photography for Samsung Galaxy S24
- Contributed to the development and commercialization of Under Display Camera for Samsung Galaxy Z Fold4
- Leading university research collaborations with Rice University

Postdoctoral Research Associate *July 2021 – April 2022*
Los Alamos National Laboratory, NM, USA

- Material Identification from Radiographs and Intrinsic Radiation
- Learning Noise-robust Features for Dynamic Reconstruction

Research Aide *July 2019 – September 2019*
Advanced Photon Source, Argonne National Laboratory, IL, USA

- *Coded Exposure for High Speed X-ray CT Imaging*: Developed CodEx, a synergistic combination of coded acquisition and reconstruction for high speed tomographic imaging

Graduate Research Assistant *January 2015 – June 2021*
Purdue University, IN, USA

- *Multi-Slice Fusion for 4D X-ray CT Reconstruction*: Developed a novel method that fuses multiple low-D Convolutional Neural Networks to implement a 4D image prior
- *Denoising Short-exposure Dynamic Radiographs*: Developed a deep-learning based transfer-learning method for denoising short-exposure dynamic radiographs
- *Multi-Domain Weighing for Metal Artifact Reduction in X-ray CT*: Developed a novel data and image domain weighing for reducing metal artifacts in X-ray CT
- *Multi-Orientation Fusion CT Reconstruction*: Developed a modular method to fuse information from multiple CT scans at different orientations to produce a joint reconstruction
- *Cone-beam X-ray CT Reconstruction of Additively Manufactured Parts*: Developed a high fidelity CT reconstruction method using scatter correction for defect detection in additively manufactured parts
- *Detection and Localization of Neurons in Fluorescence Microscopy*: Developed a novel neuron detection method to improve frame-rate of fluorescence microscopy imaging by selectively scanning neuron locations

Undergraduate Researcher *January 2014 – August 2014*
Indian Institute of Technology, Kharagpur, India

- *Efficient Wideband Spectrum Sensing for Cognitive Radio*: Developed a novel spectrum estimation algorithm for spectrum sensing in cognitive radio

Research Intern *May 2013 – July 2013*
University of Toronto, ON, Canada

- Detection of Human Thought from EEG (Electroencephalograph) Signals

Selected Publications

Majee, S., Ebenezer, J.P., Venkataramanan, A.K., Liu, W., Balke, T., Nadir, Z., Chandran, S., Lee, S.J. and Sheikh, H.R., 2026. DRIFT: Deep Restoration, ISP Fusion, and Tone-mapping. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR).*

Liang, H., Ge, Z., **Majee, S.**, Li, J., Veeraraghavan, A. and Balakrishnan, G., 2026. SplatShot: 3D Face Avatar Generation from a Single Unconstrained Photo. *arXiv preprint arXiv:2606.01493*.

Liang, H., Ge, Z., **Majee, S.**, Tiwari, A., Godaliyadda, G.M.D., Veeraraghavan, A. and Balakrishnan, G., 2025. FastAvatar: Instant 3D Gaussian Splatting for Faces from Single Unconstrained Poses. *arXiv preprint arXiv:2508.18389*.

Zhang, M., **Majee, S.**, Wang, C., Lee, S.J. and Sheikh, H., 2024. CoDISP: Exploring Compressed Domain Camera ISP with RGB-guided Encoder. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (pp. 5878–5888).

Yang, D., Kemp, C.A., **Majee, S.**, Buzzard, G.T. and Bouman, C.A., 2024, October. Pixel-weighted multi-pose fusion for metal artifact reduction in x-ray computed tomography. In *2024 IEEE 26th International Workshop on Multimedia Signal Processing (MMSP)* (pp. 1–6). IEEE.

Balke, T., Davis Rivera, F., Garcia-Cardona, C., **Majee, S.**, McCann, M.T., Pfister, L. and Wohlberg, B.E., 2022. Scientific Computational Imaging COde (SCICO). *Journal of Open Source Software*, 7(LA-UR-22-28555).

Majee, S., Aslan, S., Gursoy, D. and Bouman, C.A., 2022. CodEx: a modular framework for joint temporal de-blurring and tomographic reconstruction. *IEEE Transactions on Computational Imaging*, 8, pp. 666–678.

Majee, S., Balke, T., Kemp, C.A., Buzzard, G.T. and Bouman, C.A., 2021. Multi-slice fusion for sparse-view and limited-angle 4D CT reconstruction. *IEEE Transactions on Computational Imaging*, 7, pp. 448–462.

Majee, S., Balke, T., Kemp, C.A., Buzzard, G.T. and Bouman, C.A., 2019, May. 4D X-ray CT reconstruction using multi-slice fusion. In *2019 IEEE International Conference on Computational Photography (ICCP)* (pp. 1–8). IEEE.

Balke, T., **Majee, S.**, Buzzard, G.T., Poveromo, S., Howard, P., Groeber, M.A., McClure, J. and Bouman, C.A., 2018. Separable models for cone-beam MBIR reconstruction. *Electronic Imaging*, 2018(15), pp. 181-1.

Majee, S., Ye, D.H., Buzzard, G.T. and Bouman, C.A., 2017. A model based neuron detection approach using sparse location priors. *Electronic Imaging*, 2017(17), pp. 10–17.

Majee, S., Ray, P. and Cheng, Q., 2015, November. Efficient wideband spectrum sensing using random projection. In *2015 49th Asilomar Conference on Signals, Systems and Computers* (pp. 141–145). IEEE.

Selected Patents

Majee, S., Godaliyadda, G.M.D., Lee, J.S., Sheikh, H.R. and Polley, M.O., 2026. Interactive Selfie Panorama Capture and Multi-Perspective Undistorted Selfie Generation. *US Patent App.* 19/251,588.

Majee, S., Lee, J.S. and Sheikh, H.R., 2026. Image Haze Reduction for Backlit Scenes. *US Patent App.* 18/929,130.

Majee, S., Madhusudanarao, P.C., Nadir, Z., Lee, J.S. and Sheikh, H.R., 2025. Tone Consistency for Camera Imaging. *US Patent App.* 18/952,715.

Zhang, M., **Majee, S.**, Lee, J.S. and Sheikh, H.R., 2025. Compressed Domain Artificial Intelligence for Image Signal Processing. *US Patent App.* 18/816,846.

Godaliyadda, G.M.D., Nadir, Z., **Majee, S.**, Glotzbach, J.W., Lee, J.S. and Sheikh, H.R., 2025. Image Burst Editing Based on Natural Language Processing Input. *US Patent App.* 18/532,920.

Open Source Software

- **SVMBIR** — Super-Voxel Model Based Iterative Reconstruction. github.com/cabouman/svmbir
- **MBIRCONE** — Model-Based Iterative Reconstruction for cone-beam CT. github.com/cabouman/mbircone
- **SCICO** — Scientific Computational Imaging COde. github.com/lanl/scico

Invited Talks

Electronic Imaging	*January 2023*
Samsung Research America	*January 2022*
Oak Ridge National Laboratory	*March 2021*
Los Alamos National Laboratory	*February 2021*
Dept. of Energy Light-source Tomography Coordination Meeting	*February 2021*
Electronic Imaging	*January 2021*
Electronic Imaging	*January 2019*

Professional Volunteering

Paper Reviewer

- IEEE Transactions on Image Processing
- IEEE Transactions on Computational Imaging
- IEEE International Conference on Image Processing
- International Journal of Medical Physics Research and Practice
- Journal of Nondestructive Evaluation

Teaching Experience

Teaching Assistant *Fall 2014, Spring 2016, Spring 2017*
Purdue University

- Linear Circuits Analysis (ECE 201)
- Digital Image Processing I (ECE 637)

Scholarships & Awards

- Received president's award for contributions in High-Mega-Pixel AI Multi-Frame Processing 2024
- Received CTO award for contributions in AI Multi-frame Zoom for Galaxy S24 Ultra 2024
- Awarded MITACS Globalink Research Fellowship 2013
- Awarded the Jagadis Bose National Science Talent Search Scholarship 2010–2014
- Ranked 1 in the WBJEE Engineering Entrance Examination 2010

References

Available upon request